INTEGRATING GIS, R, AND A DIGITAL SEWAGE CAPACITY MODEL FOR FLOOD RECONSTRUCTION AND SIMULATION: INSIGHTS FROM THE BROOKLYN OCTOBER 2025 FLOOD

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Abstract:

In recent years, Brooklyn has experienced several floods that suggest the borough's stormwater infrastructure is insufficient. However, this flooding could also be caused by the landscape, so it's important to evaluate the primary cause of this issue. The study area was determined by analyzing background signage in videos and images of the NYC October 2025 flood and geolocating them in GoogleEarth. Using GIS data sets to measure the length of sewer pipes and sewer design standards from the NYC Department of Environmental Protection, the capacity of the sewage system was calculated and sea-level rise flooding was simulated. The simulations showed that the flooding patterns do not match that of a landscape flood, and that floods occurred even in areas with larger sewage capacity like on major avenues. Sewage capacity was relatively consistent across the borough, meaning capacity is not a significant issue. This then suggests flooding in multiple study points was not caused by low-lying elevation nor insufficient capacity, but by clogging of particular areas. However, data values were estimated and rainfall flooding simulation was not implemented, yet. By analyzing these flood causes, this study could help identify courses of action to reduce future flood events in Brooklyn and provide a method for other cities to analyze the causes of their own flood-prone areas.

Background Information:

- October 30, 2025, 1.83 inches were recorded in Central Park [7]. Many people took to social media to share videos of knee-deep flooding across Brooklyn during this event.
- Urban flooding is influenced not only by rainfall intensity but also by drainage system performance and localized infrastructure constraints [3, 6]. The New York City sewer system is designed to handle 1.75 inches of rain per hour [9].
- A previous study in 2024 [4] analyzed a significant September 2023 flood using flood video analysis and GIS-based methodologies. Other studies have used LiDAR-derived Digital Elevation Models (DEMs) and volumetric simulations to reconstruct flood events and evaluate inundation patterns [1, 2, 5].
- Using similar methods, this study aims to determine whether Brooklyn's floods are primarily driven by landscape conditions or sewer system behavior, recognizing that flood processes are influenced by both topographic variability and infrastructure limitations such as hydraulic connectivity and flow constraints [6, 12].

Method/Materials:

Tools: Google Earth Pro, Global Mapper (GIS), and RStudio

- Initial data collection was done using social media sites such as YouTube and TikTok contained resident-posted videos of the flood (Figure 1). These videos were analyzed for their location and height of water, and then tabulated (Table 1).
- Tabulated data was displayed as points over Brooklyn. A perimeter was then made to encompass these points as well as areas nearby.
- The perimeter from Google Earth was turned into a grid, and a road map of Brooklyn was imported in and divided amongst this grid. This was then exported to RStudio.
- In RStudio, the road length values were turned into volume values with the key assumption that each road had an equal length of pipe running parallel to it underground, a necessary assumption given the complexity of the NYC sewer system (Figure 2). Each cell now has a volume value.
- The sewage capacity values were turned into elevation values. This allowed for them to be displayed as a contour map, the desired method of viewing the study area's capacity.

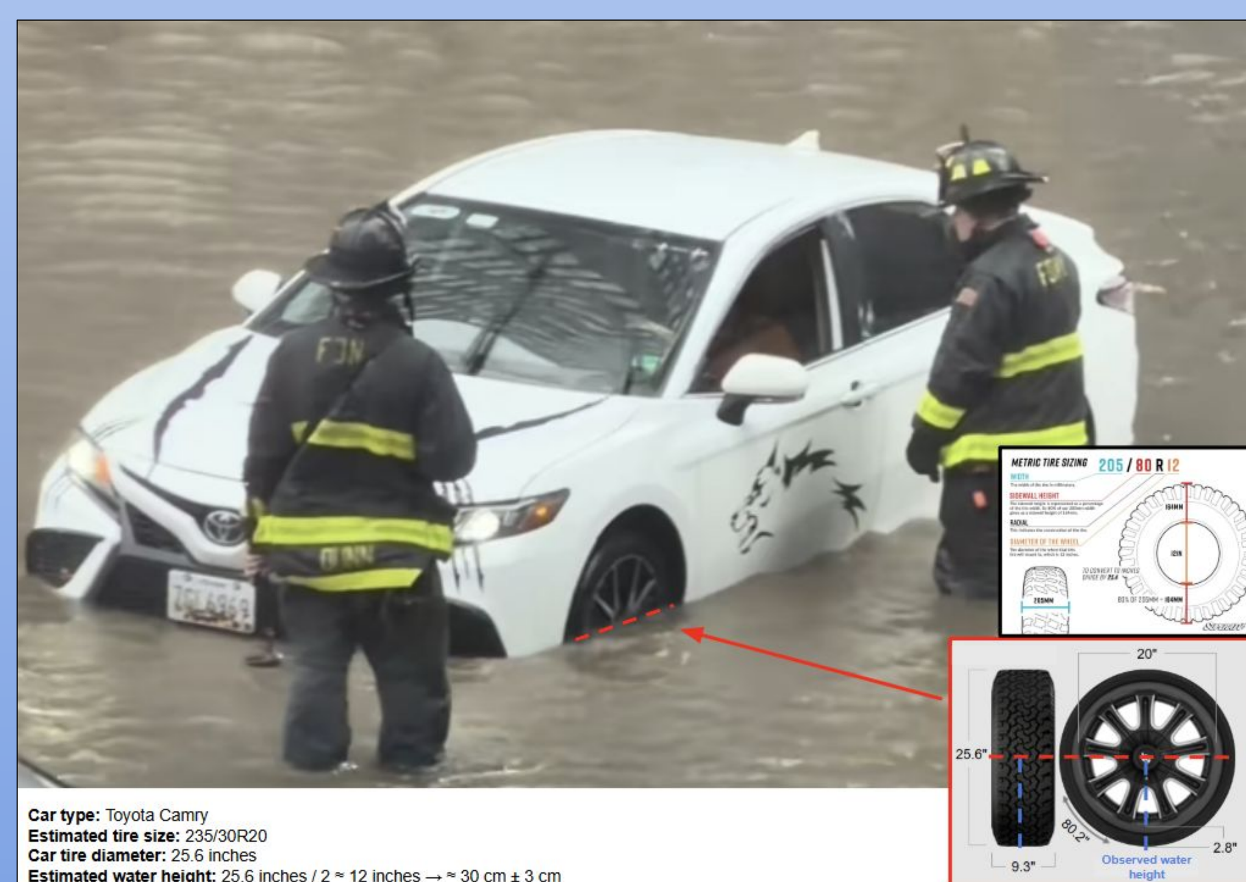

Figure 1: Screenshot of one of the flood videos used for analysis [14], and graphic of tire models used for flood depth estimate [10, 11, 13].

Map #	Location (Intersection, Address, Neighborhood)	Coordinates (Longitude, Latitude)	Water Height Inferred (cm) & Uncertainty (±)	Link to Source and Timestamp/Note
0	**4024 Farragut Rd** East Flatbush	40°38'11.91" N 73°56'15.62" W	18 ± 2 cm	https://youtu.be/1ywSNVWsQlI?si=N3xj78fEb3VLCKR4 (00:25-00:35)
1	**1523 Avenue M** Midwood	40°37'06.48" N 73°57'33.65" W	12 ± 3 cm	https://www.youtube.com/watch?v=jgXVxQ3wo1s (00:21-00:27)
2	**372 4th Avenue** Park Slope	40°40'22.68" N 73°59'12.52" W	18 ± 2 cm	https://www.tiktok.com/@sosyy122/video/7567397149529378103?is_from_webapp=1&sender_device=pc&web_id=7608235670612231711 (00:25-00:40)
3	**1933 Flatbush Ave** Flatbush	40°37'22.23" N 73°56'11.58" W	8 ± 3 cm	https://www.tiktok.com/@unmainstreamed/video/7567833535868964151?is_from_webapp=1&sender_device=pc&web_id=7608235670612231711 (00:08-00:46)
4	**4th Ave, 5th Street*** Park Slope	40°40'21.93" N 73°59'12.02" W	18 ± 2 cm	https://www.youtube.com/shorts/TjY5qH109sU (full video)
5	**6th Ave & Pacific St*** Prospect Heights	40°40'53.96" N 73°58'26.12" W	28 ± 3 cm	https://www.tiktok.com/@squatsnsplits92/video/7567107534989839646?_r=1&_t=ZP-941nrvtOBdn (full video)

Table 1: Excerpt of locations of observed flooding across Brooklyn during the October 2025 storm event, including geographic coordinates, estimated water depths with associated uncertainties, and corresponding source media used for reconstruction. See paper for remaining locations.

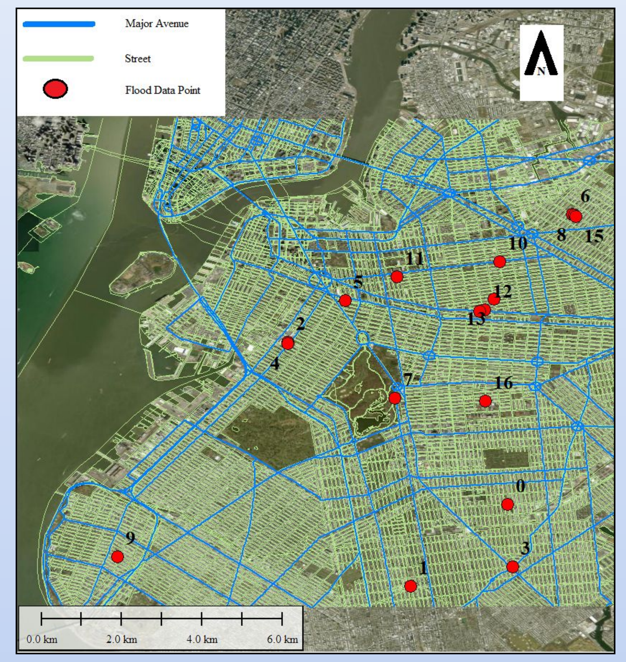

Figure 2: All roads in Brooklyn that fall within the specified study area. Major avenues are specified blue, and streets are specified in green. Roads/bridges in the small Manhattan portion are not considered in mapping analysis. Source: [8]

Results:

Figure 3 details sewage capacity in the streets and major avenues of Brooklyn. Notably, high capacity areas like Point 7 in Figure 3 are still subject to flooding. The distribution of sewage capacity across Brooklyn is relatively uniform, with expectedly more being present on major avenues (Figure 3). The landscape flood simulation (Figure 4a, 4b, 4c) covered many points, but not all of them. At the end of this simulation, close to half the points still remain outside the flood zone.

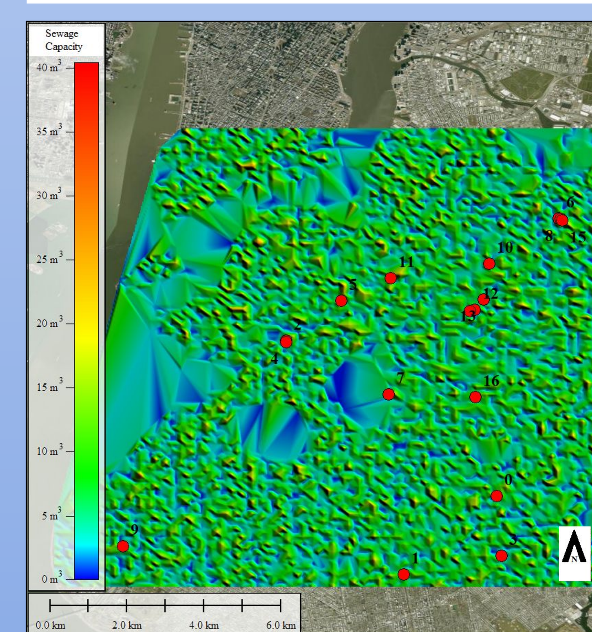

Figure 3: Sewage capacity of Brooklyn streets represented as a colored map, referred to as a digital sewage capacity model. Large bluish-green voids can be seen, these represent parks and waterways.

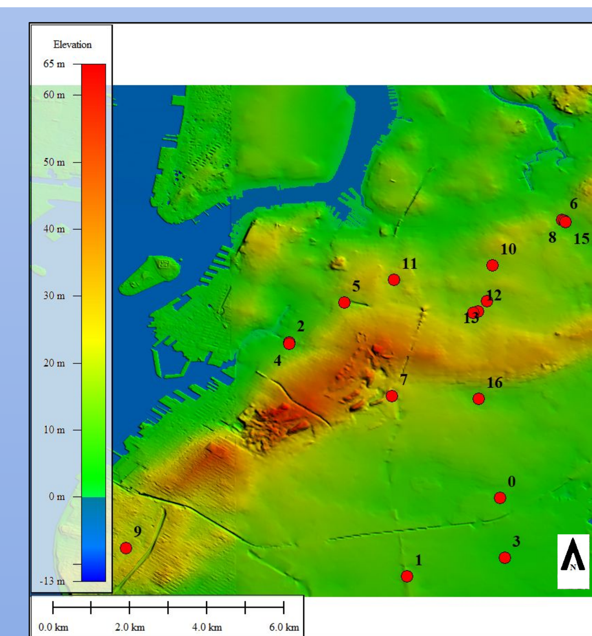

Figure 4a: Simulation of a landscape flood, with this view being the default DEM view of Brooklyn. Note the high elevation areas and the notable amount of data points near them. Sea level is at default currently.

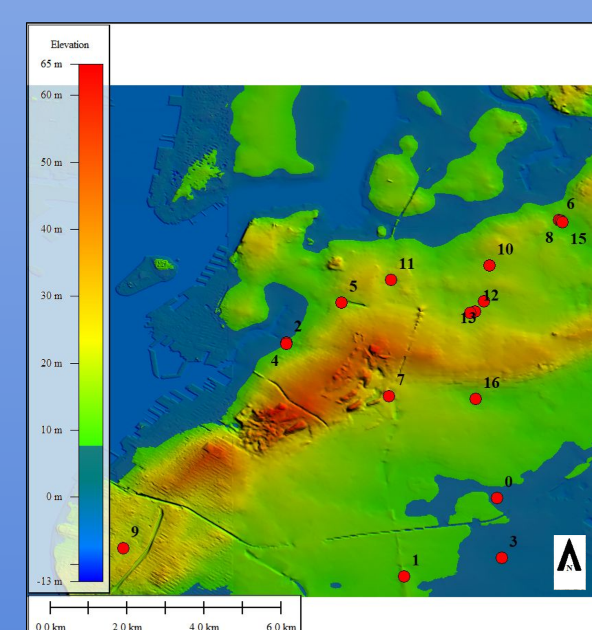

Figure 4b: Simulation of a landscape flood, but the water level has risen by 8 meters. Four data points have become submerged.

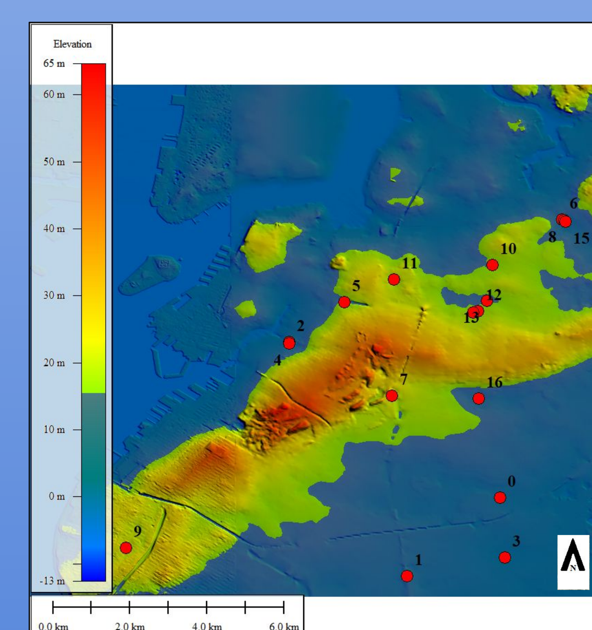

Figure 4c: Simulation of a landscape flood, but the water level has risen by 16 meters. Most, not all, of the flooded points have been overlapped by the water level.

Discussion:

- The landscape-based flood simulation indicated that even under an extreme and unrealistic scenario, the modeled flood extent did not match the observed flooded locations identified earlier in this study.
- The sewage discharge rate to waterways/waste treatment facilities was likely adequate for this flood in particular, suggesting that sewer backups could be the issue instead.
- A significant limitation to note is Global Mapper's inability to simulate a non-landscape flood, especially with uneven terrain not in a "bathtub" shape, such as Brooklyn.

Conclusions:

- The results of this study suggest that sewer system limitations likely played a major role in flooding during this event.
- Further steps could be taken to get a sewage capacity model that accounts for the specifics of the NYC sewer system better and additional systems connected to the sewers like wastewater treatment plants.

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Credit Authorship Contribution Statement:

Persaud, D.: Data Collection, Global Mapper, programming in R (RStudio), Figures, Writing: Abstract, Methodology, Results, Discussion, Conclusion; Moore, O.: Data Collection, Research, Writing: Editing, Abstract, Introduction, Methodology, Discussion, Conclusion; Mercaldo, D.: Data Collection, Google Earth, RStudio, Microsoft Excel, References, Editing, Figures, Writing: Abstract, Introduction, Methodology, Discussion, Conclusion; Marsellos, A.E.: Supervision, Editing, RStudio

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More references available in full report.